The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 June 2004.

INCH-POUND

MIL-PRF-19500/632B
16 March 2004
SUPERSEDING
MIL-PRF-19500/632A
31 March 1998

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, N-CHANNEL
SILICON, TYPES 2N7399, 2N7400, 2N7401, AND 2N7402,
JANSD AND JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event characterization), power transistor. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, (similar to TO-257).

* 1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1) T_C =+25°C	P_T T_A =+25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} (2) (3) T_C =+25°C	I_{D2} T_C = +100°C	I_S	I_{DM}	T_J and T_{STG}	V_{ISO} 70,000 ft. altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7399	50	20	100	100	±20	11.0	7.0	11.0	33	-55	N/A
2N7400			200	200		8.0	5.0	8.0	24	to	N/A
2N7401			250	250		6.0	4.0	6.0	18	+150	250
2N7402			500	500		3.0	2.0	3.0	9		500

(1) Derate linearly 0.4 W/°C for T_C > +25°C;

(2) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(3) See figure 2, maximum drain current graphs.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://www.dodssp.daps.mil/.

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc	Max I_{DSS1} V_{GS} = 0 V_{GS} = 80% of rated V_{DS}	Max $r_{DS(on)}$ (1) V_{GS} = 12V		$R_{\theta JC}$ Max	$I_{AS} = I_{DM}$
				T_J = +25°C at I_{D2}	T_J = +125°C at I_{D2}		
	V dc	V dc Min Max	µA dc	Ω	Ω	°C/W	A (pk)
2N7399	100	1.5 4.0	25	0.21	0.351	2.50	33
2N7400	200			0.44	0.744		24
2N7401	250			0.60	1.08		18
2N7402	500			2.70	5.37		9

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

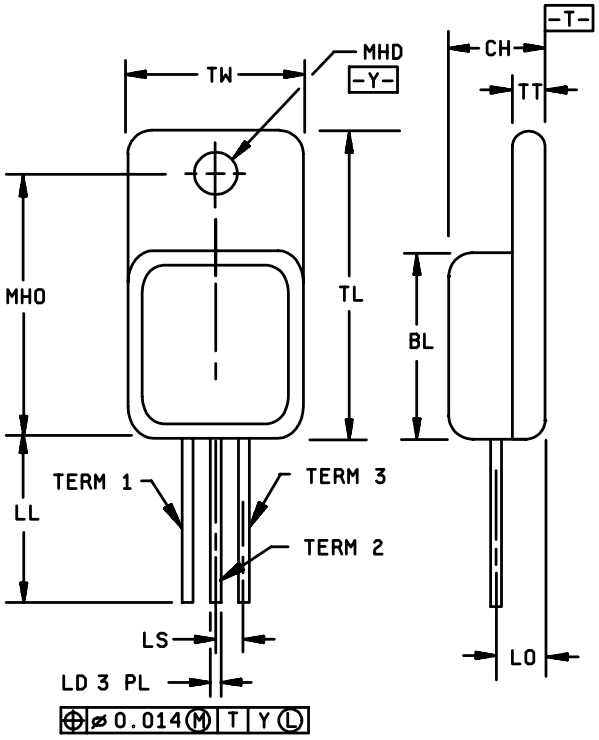
* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://www.dodssp.daps.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.405	.425	10.29	10.80
CH	.190	.200	4.83	5.08
LD	.025	.035	0.64	0.89
LL	.600	.650	15.24	16.51
LO	.120 BSC		3.05 BSC	
LS	.100 TYP		2.54 TYP	
MHD	.140	.150	3.56	3.81
MHO	.522	.542	13.26	13.77
TL	.645	.665	16.38	16.89
TT	.035	.045	0.89	1.14
TW	.410	.420	10.41	10.67
Term 1	Gate			
Term 2	Drain			
Term 3	Source			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4. Die to base is BeO isolated, terminals to case ceramic (AL_2O_3) isolated.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Physical dimensions (similar to TO-257).

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

* 3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

nC.....nano coulomb.
I_{AS}.....Rated avalanche current, non-repetitive.

* 3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1.

* 3.4.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq$ or 100 kΩ, whenever bias voltage is applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

* 3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

* 3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor but shall be retained on the initial container.

* 3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

* 4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein shall be performed by the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANS only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500) (1) (2)	Measurement
	JANS
(3)	Gate stress test (see 4.3.1)
(3)	Method 3470 of MIL-STD-750, (see 4.3.2)
(3) 3c	Method 3161 of MIL-STD-750, (see 4.3.3)
7	Optional.
9	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, subgroup 2 of table I herein.
10	Method 1042 of MIL-STD-750, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.
12	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$ = ±20 percent of initial value.
14	Required.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime before screen 9.

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* 4.3.1 Gate stress test. Apply V_{GS} = 30 V minimum for t = 250 µs minimum.

* 4.3.2 Single pulse avalanche energy (E_{AS}).

a. $I_{AS} = I_{DM}$.

b. L = 0.1 mH .

c. $E_{AS} = 1/2\ LI_{AS}^2$.

d. V_{DD} = 50 V to 150 V dc.

e. Initial junction temperature = +25°C, -5°C, +10°C.

* 4.3.3 Thermal impedance (ΔV_{SD} measurement). The ΔV_{SD} measurement shall be performed in accordance with method 3161 of MIL-STD-750. The ΔV_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 3) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

a. Measuring current (I_M) .....10 mA.

b. Heating time (t_H) .....100 ms.

c. Measurement time delay (t_{MD}) .....30 - 60 µs.

d. Sample window time (t_{SW}) .....10 µs maximum.

* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500, and as follows. End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles
B3	2077	SEM
B4	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink nor or forced-air cooling on the device shall be permitted during the on cycle. t_{ON} = 30 seconds minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175°C; t = 120 hours, minimum.
B5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175°C; t = 24 hours.
B5	2037	Bond strength, test condition A
B6	3161	Thermal resistance, see 4.5.2.

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* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

Subgroup	Method	Condition
C2	2036	Terminal strength, test condition A, weight = 10 pounds., t = 15 seconds.
C5	3161	See 4.5.2.
C6	1042	Test condition D, 6,000 cycles; 1 cycle = 30 seconds minimum.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF-19500 and table II herein.

4.4.4.1 Design parameters. Not tested on a per lot basis. Design shall be such that the devices shall be capable of meeting the requirements herein.

* 4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC}$ = 2.5 ^{o}C/W. The following parameters shall apply:

a. Measuring current(I_M) ..... 10 mA.

b. Drain heating current (I_H) ..... 1 A.

c. Heating time (t_H) ..... Steady-state (see method 3161 of MIL-STD-750).

d. Drain-source heating voltage (V_H) ... 25 V.

e. Measurement time delay (t_{MD}) ..... 30 to 60 µs.

f. Sample window time (t_{SW}) ..... 10 µs maximum.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, drain to source	3407	Bias condition C V_{GS} = 0 V, I_D = 1.0 mA dc	$V_{(BR)DSS}$			
2N7399				100		V dc
2N7400				200		V dc
2N7401				250		V dc
2N7402				500		V dc
Gate to source voltage (threshold)	3403	$V_{DG} \geq V_{GS}$,	$V_{GS(th)1}$	1.5	4.0	V dc
Gate current	3411	Bias condition C, V_{GS} = +20 V dc, V_{DS} = 0	I_{GSSF1}		+100	nA dc
Gate current	3411	Bias condition C, V_{GS} = -20 V dc, V_{DS} = 0	I_{GSSR1}		-100	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V dc, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	Bias condition A, V_{GS} = 12 V dc, I_D = I_{D1}, pulsed (see 4.5.1)	$r_{DS(ON)1}$			
2N7399					0.21	Ω
2N7400					0.44	Ω
2N7401					0.60	Ω
2N7402					2.70	Ω
Static drain to source on-state voltage	3405	Bias condition A, V_{GS} = 0 V dc, I_D = I_{D1}, pulsed (see 4.5.1)	$V_{DS(ON)1}$			
2N7399					2.43	V dc
2N7400					3.70	V dc
2N7401					3.78	V dc
2N7402					8.51	V dc
Forward voltage	4011	Pulsed (see 4.5.1), I_D = I_{D1} V_{GS} = 0 V dc	V_{SD}		1.8	V dc

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate current	3411	Bias condition C, $V_{GS} = \pm 20$ V dc, $V_{DS} = 0$ V dc,	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C, $V_{GS} = 0$ V dc, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		0.25	mA dc
Static drain to source on-state resistance	3421	V_{GS} = -10 V dc, pulsed (see 4.5.1), I_D = rated I_{D2}	$r_{DS(on)2}$			
2N7399					0.351	Ω
2N7400					0.744	Ω
2N7401					1.08	Ω
2N7402					5.37	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, $I_D = 1$ mA	$V_{GS(th)2}$	0.5		V dc
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, $I_D = 1$ mA	$V_{GS(th)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}; V_{GS} = 12 V dc; $R_G = 7.5\ \Omega$; V_{DD} = 50 percent of rated V_{DS} ($V_{DS} \leq 200$V)				
Turn-on delay time			$t_{d(on)}$			
2N7399					65	ns
2N7400					65	ns
2N7401					70	ns
2N7402					85	ns
Rise time			t_r			
2N7399					200	ns
2N7400					160	ns
2N7401					140	ns
2N7402					120	ns

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 4 - Continued						
Turn-off delay time			$t_{d(off)}$			
2N7399					130	ns
2N7400					120	ns
2N7401					120	ns
2N7402					150	ns
Fall time			t_f			
2N7399					90	ns
2N7400					90	ns
2N7401					95	ns
2N7402					85	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 4, t_p = 10 ms, V_{DS} = 80 percent of rated V_{DS}, V_{DS} ≤200 V dc max.				
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7399					42	nC
2N7400					42	nC
2N7401					41	nC
2N7402					36	nC
Gate to source charge			Q_{gs}			
2N7399					11	nC
2N7400					12	nC
2N7401					11	nC
2N7402					8.4	nC

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 7 - Continued						
Gate to drain charge			Q_{gd}			
2N7399					23	nC
2N7400					22	nC
2N7401					21	nC
2N7402					15	nC
Reverse recovery time	3473	$d_i/d_t \leq 100$ A/µs, $V_{DD} \leq 30$ V,	t_{rr}			
2N7399					280	ns
2N7400					340	ns
2N7401					420	ns
2N7402					400	ns

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/ 4/	MIL-STD-750		Symbol	Pre-irradiation limits		Post irradiation limits		Units
	Method	Conditions		Min	Max	Min	Max	
Subgroup 1								
Not applicable								
Subgroup 2		T_C = +25°C						
Steady state total dose irradiation (V_{GS} bias) 4/	1019	V_{GS} = 12 V, V_{DS} = 0 V						
Steady state total dose irradiation (V_{DS} bias)	1019	V_{GS} = 0 V, V_{DS} = 80 percent of rated V_{DS}						
Breakdown voltage drain to source	3407	V_{GS} = 0 V, I_D = 1 mA dc,	$V_{(BR)DSS}$					
2N7399		Bias condition C		100		100		V dc
2N7400				200		200		V dc
2N7401				250		250		V dc
2N7402				500		500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D =1 mA dc	$V_{GS(TH)1}$	1.5	4.0	1.5	4.0	V dc
Gate current	3411	V_{GS} = ±20 V dc, V_{DS} = 0 V, Bias condition C	I_{GSS1}		±100		±100	nA dc
Drain current	3413	V_{GS} = 0 V, V_{DS} = 80 percent of rated V_{DS}, Bias condition C	I_{DSS1}		25		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see4.5.1), I_D = I_{D2}	$r_{DS(ON)1}$					
2N7399					0.21		0.21	Ω
2N7400					0.44		0.44	Ω
2N7401					0.60		0.60	Ω
2N7402					2.70		2.70	Ω
Static drain to source on-state voltage	3405	V_{GS} = 12 V dc, condition A, Pulsed (see 4.5.1),	$V_{DS(ON)}$					
2N7399		I_D = I_{D1}			2.43		2.43	V dc
2N7400					3.70		3.70	V dc
2N7401					3.78		3.78	V dc
2N7402					8.51		8.51	V dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical specifications are for 'D' and 'R' rad levels.
3/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification utilizing the same die design.
4/ At the manufacturer's option, group D samples need not be subjected to all the screening tests, but shall be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.

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* TABLE III. Group E inspection (all quality levels) for qualification or requalification only.

Inspection 1/ 2/ 3/ 4/ 5/	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycle	1051	Condition G, 500 cycles	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8	
Subgroup 2 1/			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			3 devices, c = 0
DPA	2102		
Subgroup 4			Sample size N/A
Thermal impedance curves		Each supplier shall submit their (typical) design maximum thermal impedance curves. In addition, the optimal test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report	
Subgroup 5			5 devices c = 0
Barometric pressure test (not required for $V_{BR(DSS)} \leq$ 200 V) 2N7401 2N7402	1001	 V_{DS} = 250 V; $I_{(ISO)}$ < 0.25 mA V_{DS} = 500 V; $I_{(ISO)}$ < 0.25 mA	
Subgroup 6			3 devices
ESD	1020	Not required for devices classified as ESD class 1.	
Subgroup 8			
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476		22 devices c = 0

See footnotes at end of table.

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* TABLE III. Group E inspection (all quality levels) for qualification or requalification only - Continued.

Inspection 1/ 2/ 3/ 4/ 5/	MIL-STD-750		Qualification and large lot conformance inspection
	Method	Conditions	
Subgroup 9			3 devices c = 0 5/
Electrical measurements 3/		See table IV, steps 3 and 4	
SEE effect testing 4/	1080	See figure 5. Fluence = 3e5 ± 20 percent ions/cm^2 Flux = 5e3 to 2e4 ions/cm^{2}sec Beam energy = 260 to 300 MeV Temperature = 25°C ± 5°C (see figure 5)	
2N7399		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 100 V and V_{GS} = -20 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 100 V and V_{GS} = -10 V V_{DS} = 80 V and V_{GS} = -15 V V_{DS} = 50 V and V_{GS} = -20 V	
2N7400		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 200 V and V_{GS} = -20 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 200 V and V_{GS} = -5 V V_{DS} = 160 V and V_{GS} = -10 V V_{DS} = 100 V and V_{GS} = -15 V V_{DS} = 40 V and V_{GS} = -20 V	
2N7401		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 250 V and V_{GS} = -15 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 250 V and V_{GS} = -5 V V_{DS} = 200 V and V_{GS} = -10 V V_{DS} = 125 V and V_{GS} = -15 V V_{DS} = 50 V and V_{GS} = -20 V	
2N7402		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 500 V and V_{GS} = -15 V V_{DS} = 450 V and V_{GS} = -20 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 500 V and V_{GS} = -5 V V_{DS} = 400 V and V_{GS} = -10 V V_{DS} = 100 V and V_{GS} = -15 V	

See footnotes at end of table.

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* TABLE III. Group E inspection (all quality levels) - For qualification or re-qualification only - Continued.

Inspection 1/ 2/ 3/ 4/ 5/	MIL-STD-750		Qualification and large lot conformance inspection
	Method	Conditions	
Subgroup 9 - Continued. Electrical measurements 4/		See table IV, steps 3 and 4	

1/ A separate sample for each test may be pulled.
2/ Group E qualification of single event effect testing may be performed prior to lot formation. Wafers qualified to these group E QCI requirements may be used for any other specification utilizing the same die design.
3/ As a minimum, gate to source leakages and drain to source leakage are to be examined to verify the electrical performance of the DUT prior to and after test. At the manufacturer's option, the remaining static tests in table IV, with the exception of step 8, may be performed.
4/ Devices passing a given combination of drain and gate voltage for an LET of 36 to 40 MeV-cm^2/mg qualify the same conditions for an LET of 26 to 30 MeV-cm^2/mg.
5/ This sampling plan applies to each bias condition specified.

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* TABLE IV. Group A, B, C, and E electrical and delta measurements.

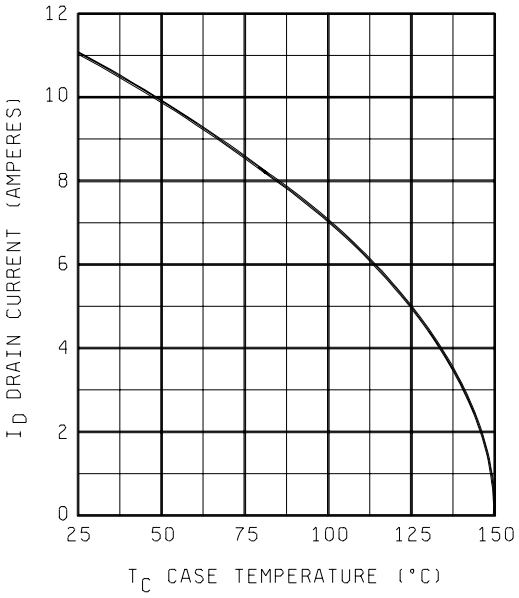
Step	Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Breakdown voltage drain to source 2N7399 2N7400 2N7401 2N7402	3407	V_{GS} = 0 V, I_D = 1 mA dc, bias condition C	$V_{(BR)DSS}$	 100 200 250 500		 V dc V dc V dc V dc
2.	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)1}$	1.5	4.0	V dc
3.	Gate current	3411	V_{GS} = +20 V and -20 V dc bias condition C, V_{DS} = 0 V	I_{GSS1}		±100	na dc
4.	Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
5.	Static drain to source on-state resistance 2N7399 2N7400 2N7401 2N7402	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)1}$		 0.21 0.44 0.60 2.70	 Ω Ω Ω Ω
6.	Static drain to source on-state voltage 2N7399 2N7400 2N7401 2N7402	3405	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D1}$	$V_{DS(ON)}$		 2.434 3.70 3.78 8.51	 V dc V dc V dc V dc
7.	Forward voltage	4011	V_{GS} = 0 V dc, condition A, pulsed (see 4.5.1; $I_D = I_{D1}$	V_{SD}		1.8	V dc
8.	Thermal impedance	3161	See 4.3.3	ΔV_{SD}		90	mV

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
- a. Subgroup 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
- b. Subgroup 4, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.
- c. Subgroup 5, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

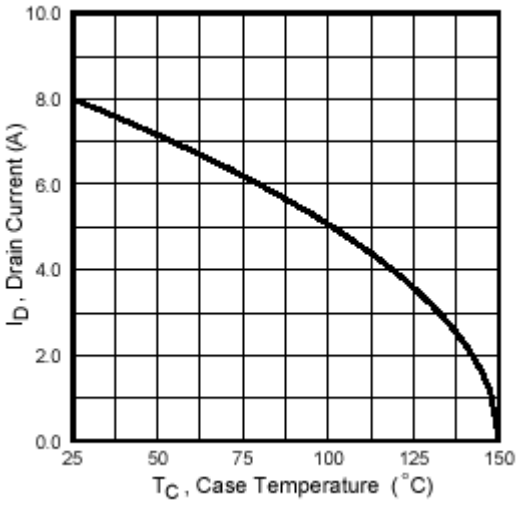
2/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
- a. Subgroup 2 and 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
- b. Subgroup 6, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

3/ The electrical measurements for table IX of MIL-PRF-19500 are as follows:
- a. Subgroups 1, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.
- b. Subgroups 2, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
- c. Subgroup 8, see table IV herein, steps 3 and 4.

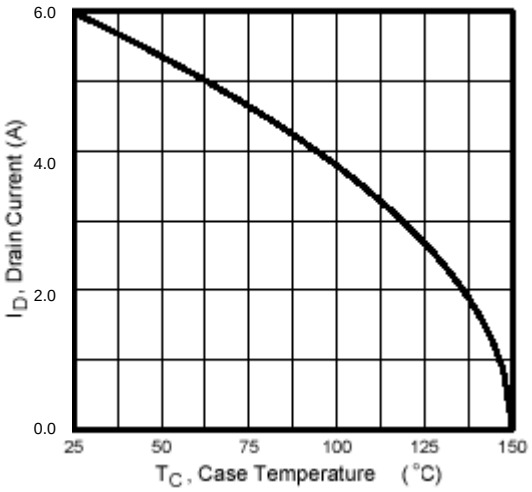
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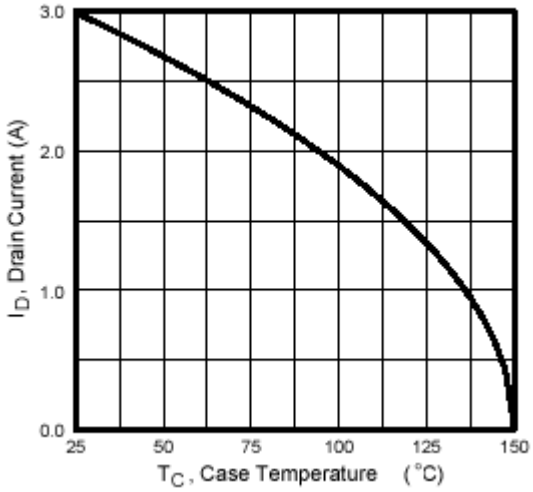
2N7399



2N7400



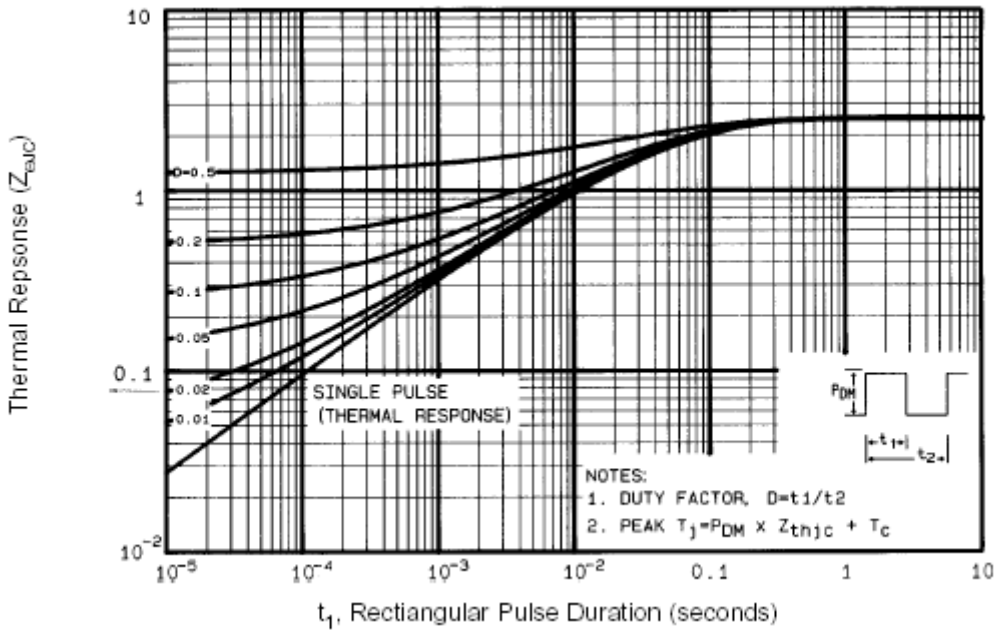
2N7401



2N7402

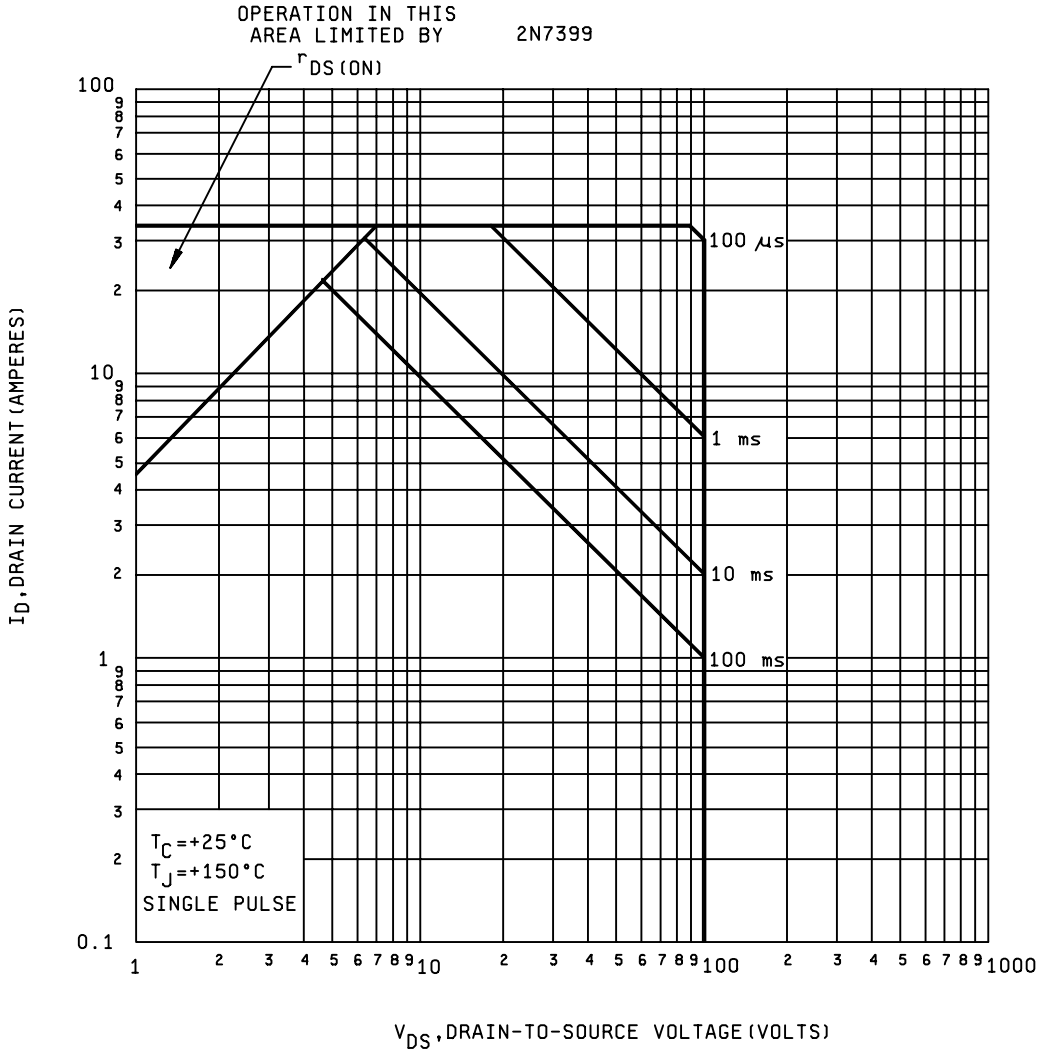
* FIGURE 2. Maximum drain current vs case temperature graphs.

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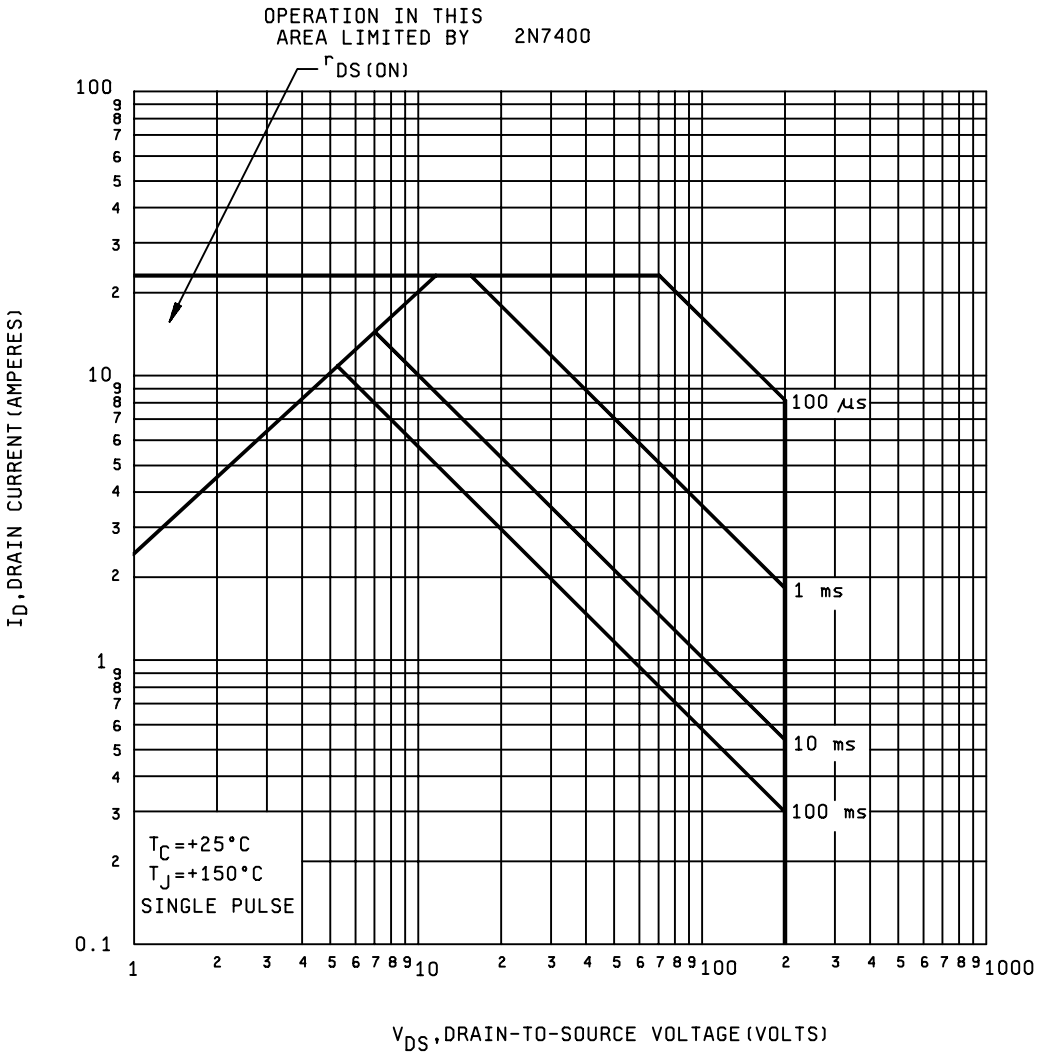
* FIGURE 3. Thermal response curves

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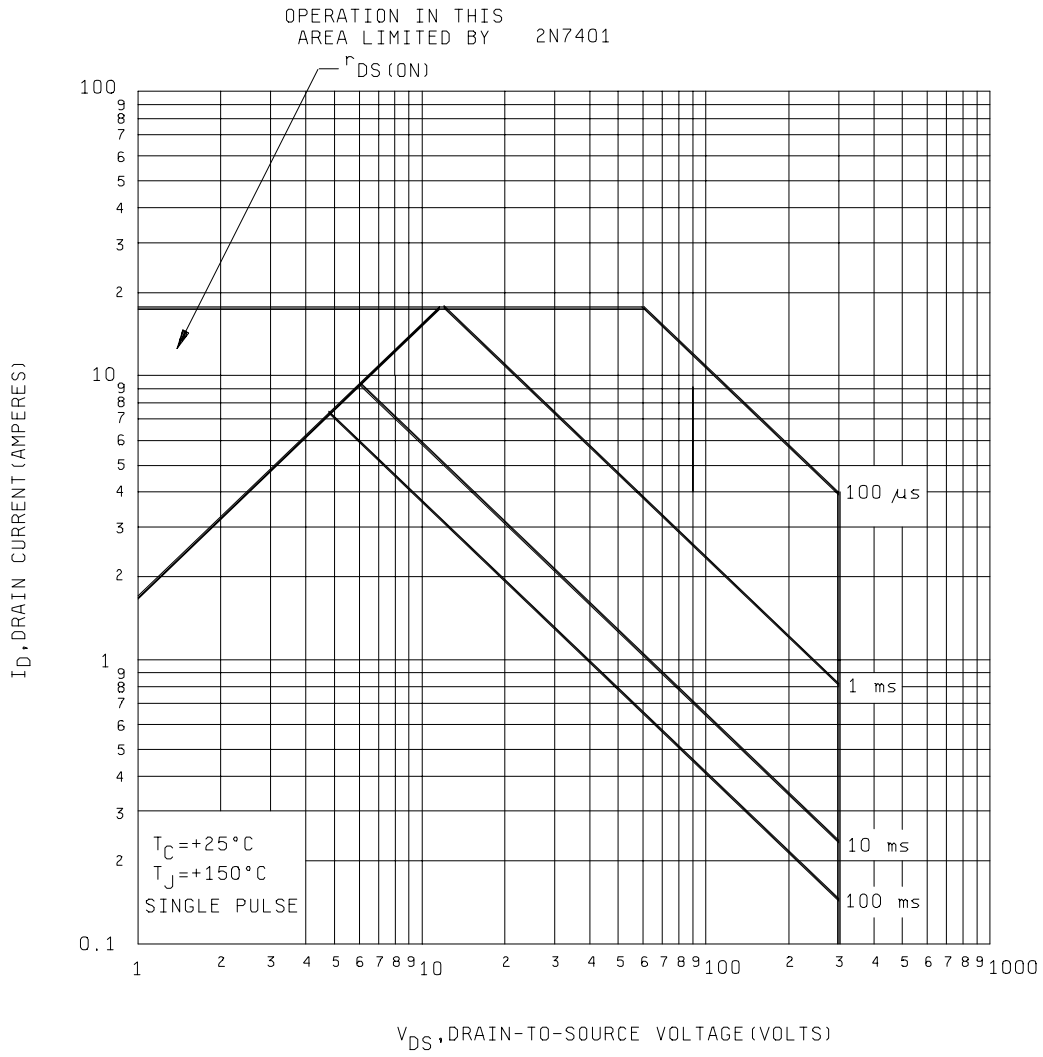
* FIGURE 4. Safe operating area graphs

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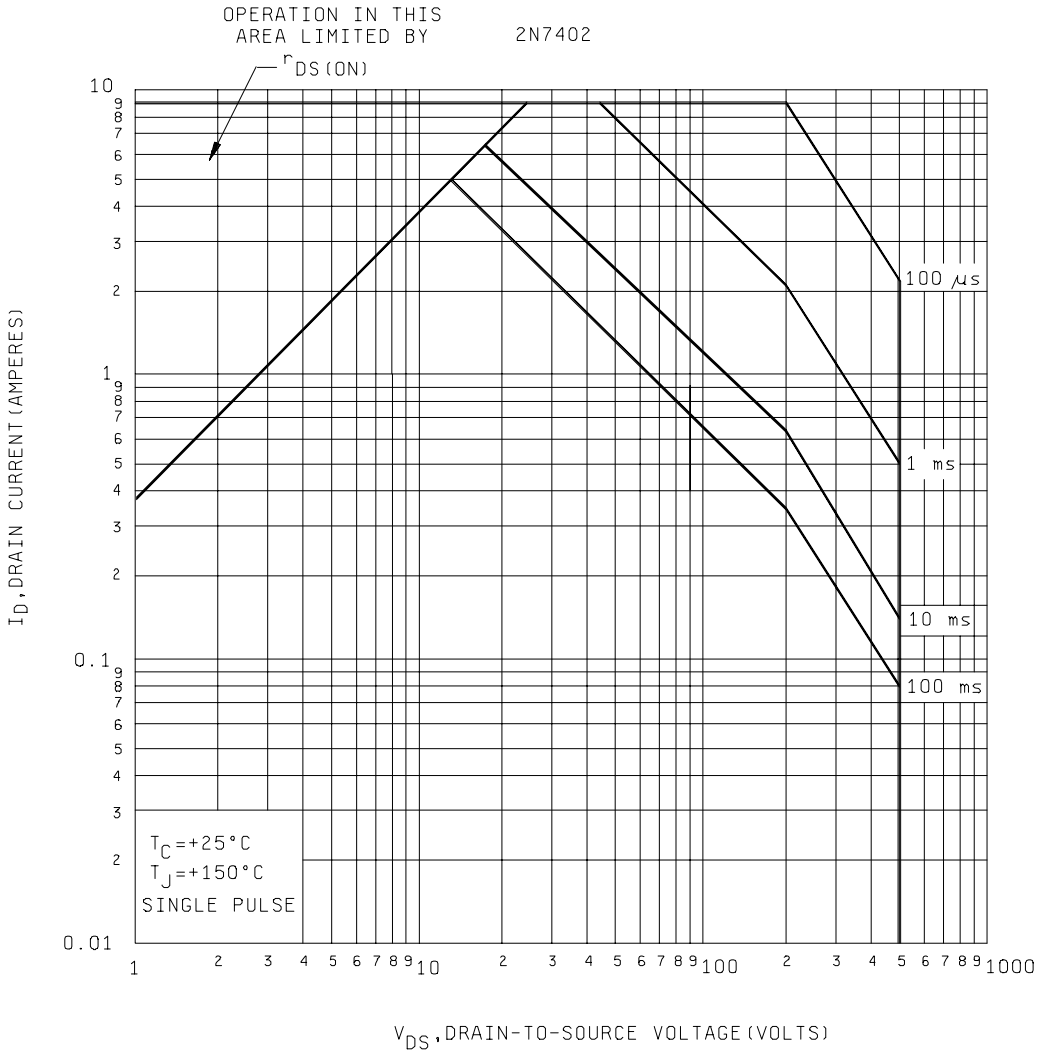
* FIGURE 4. Safe operating area graphs - Continued.

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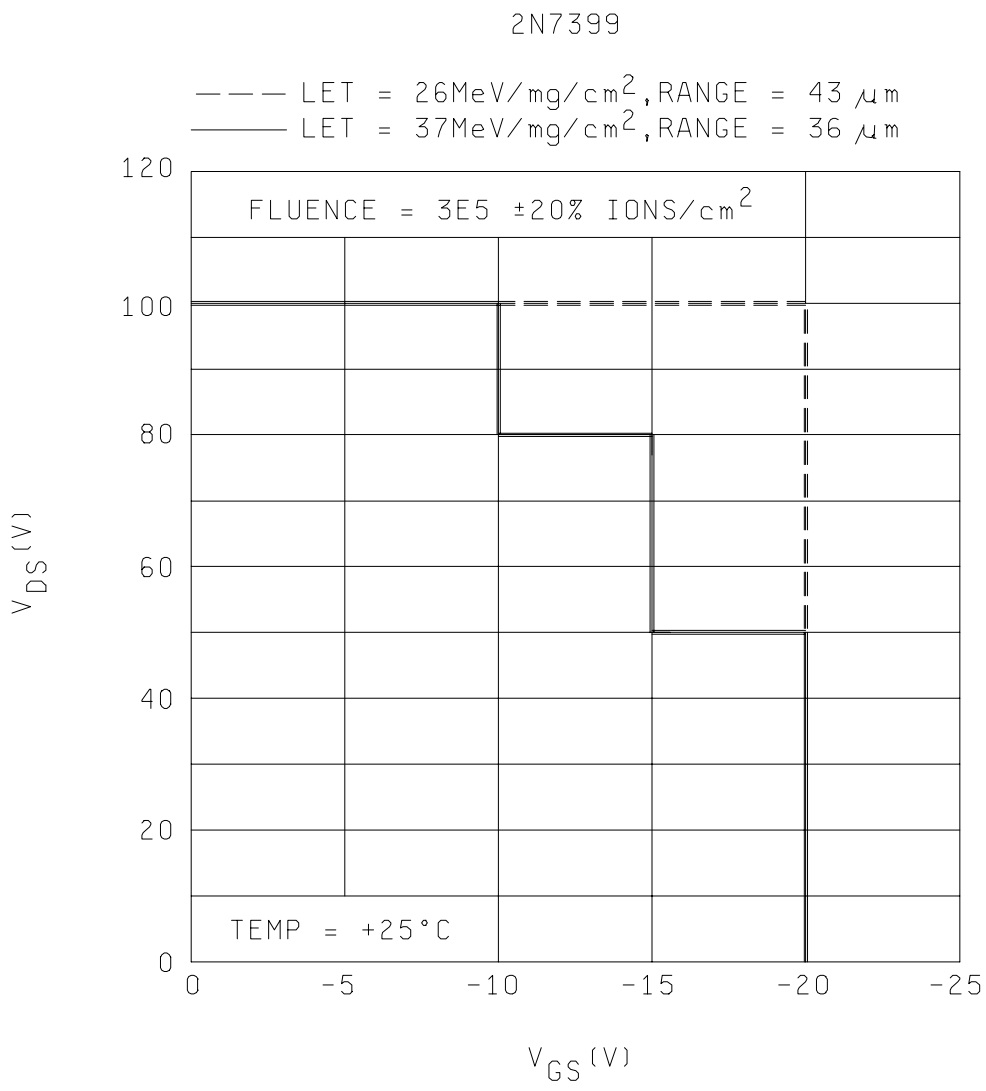
* FIGURE 4. Safe operating area graphs - Continued.

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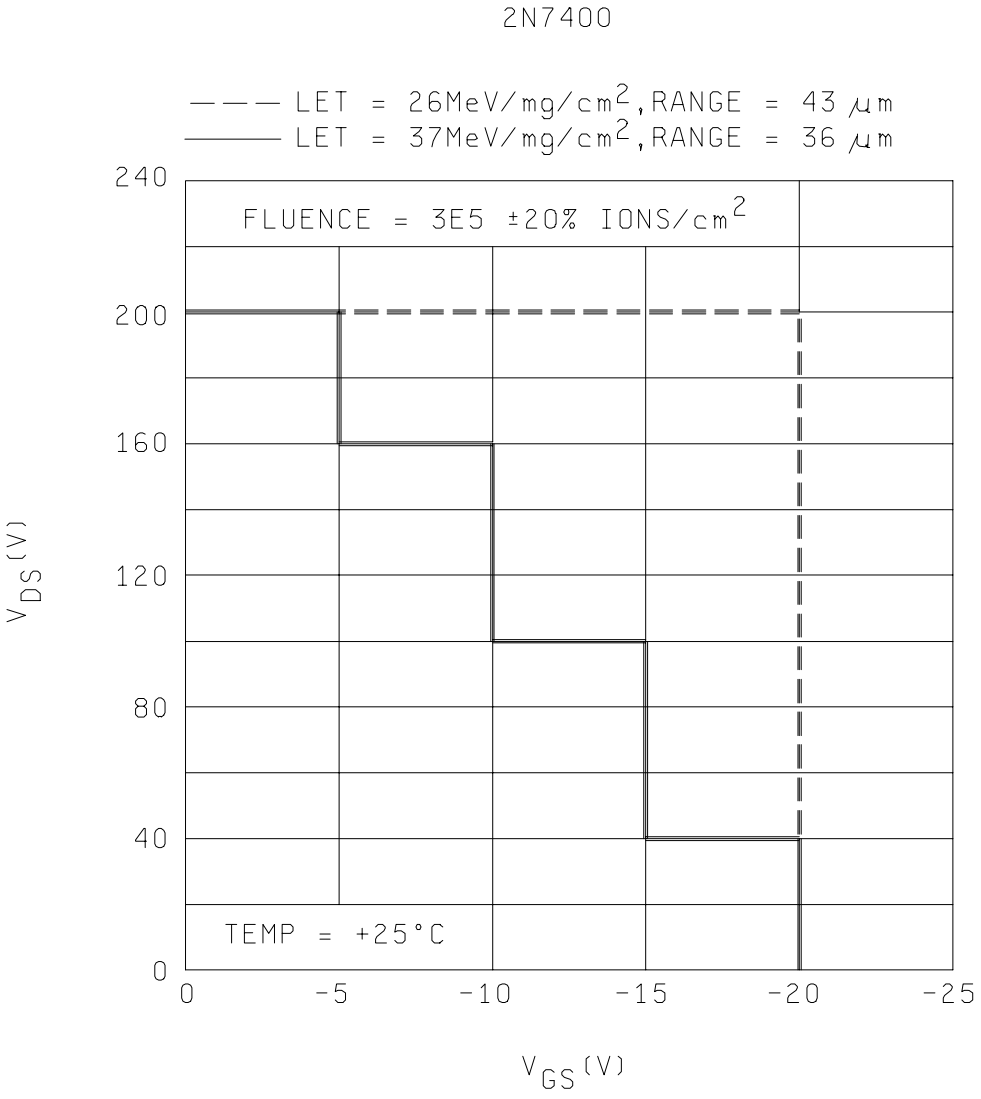
* FIGURE 4. Safe operating area graphs - Continued.

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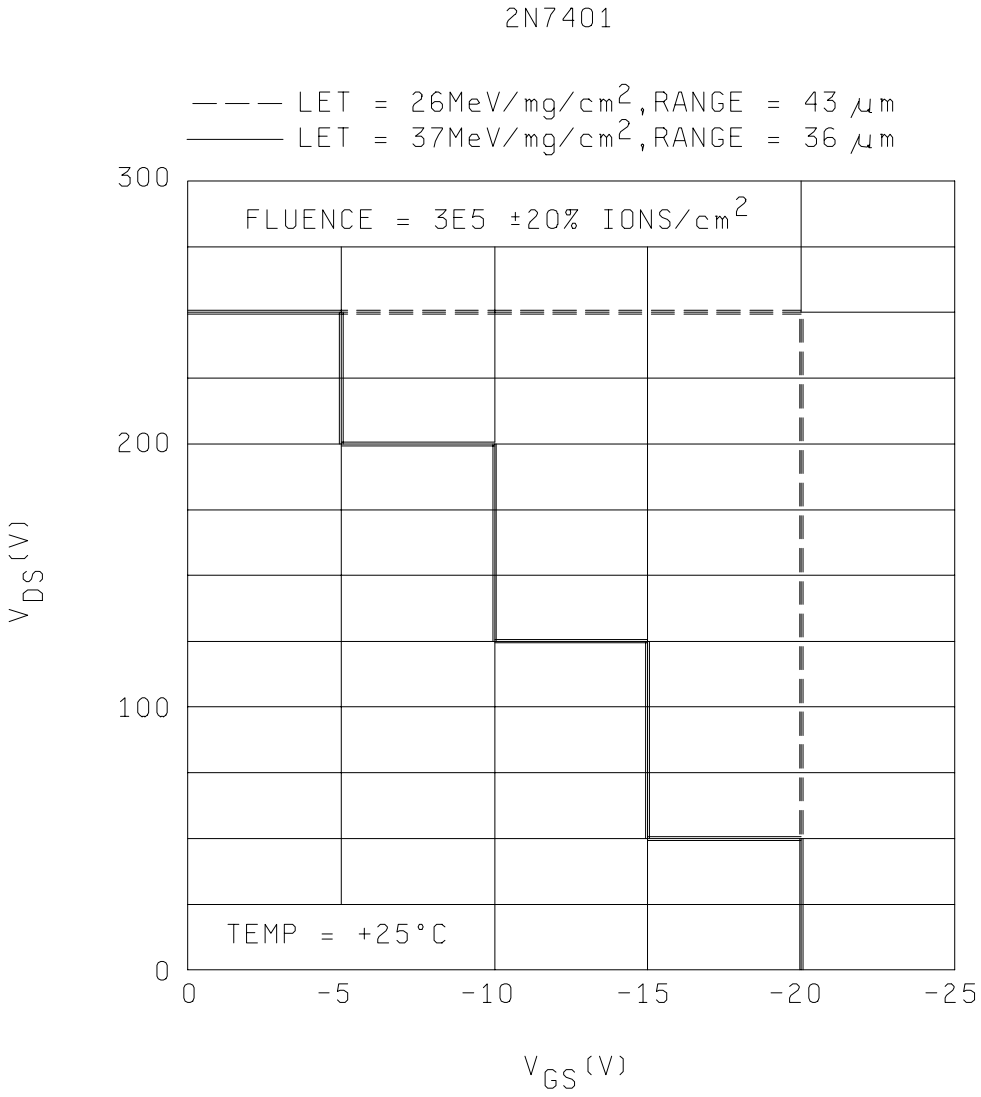
* FIGURE 5. Single event effects safe operating area graphs.

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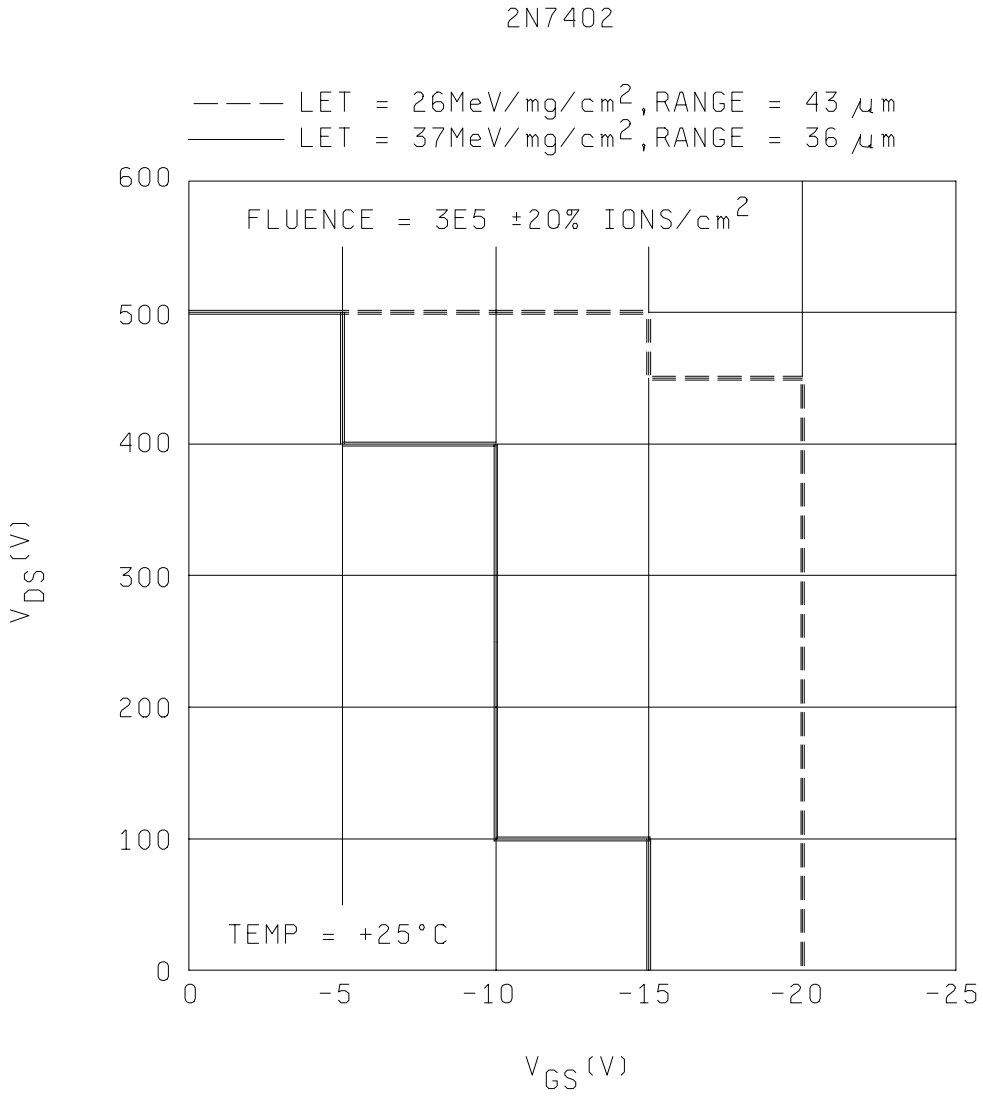
* FIGURE 5. Single event effects safe operating area graphs - Continued.

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* FIGURE 5. Single event effects safe operating area graphs - Continued.

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* FIGURE 5. Single event effects safe operating area graphs - Continued.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead material and finish may be specified (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000 or e-mail vqe.chief@dla.mil.

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* 6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PINs are subitable for the military PIN.

Generic P/N	Military P/N
SS130	2N7399
SS230	2N7400
SS234	2N7401
SS430	2N7402

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AV, MI

Preparing activity:
DLA - CC

(Project 5961-2851)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil/ .

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